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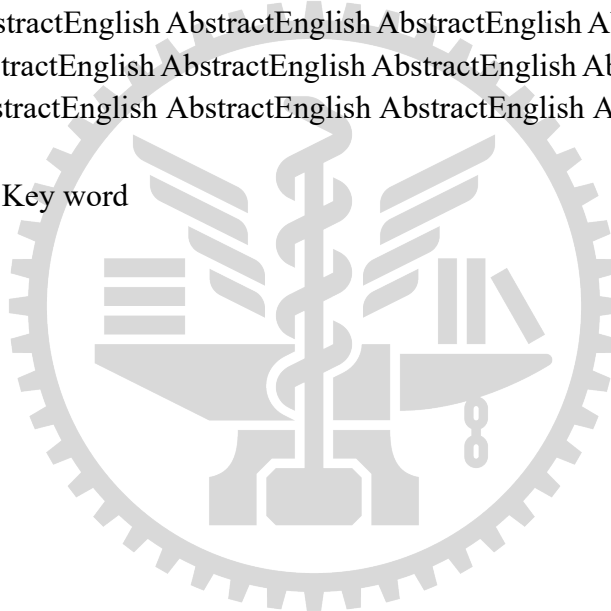
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Abstract

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Chapter 1 Introduction

1.1 Exciton in Transition Metal Dichalcogenide (TMD) Monolayer

Conventional semiconductors such as diamond, silicon, and aluminum arsenide are not suitable for the selective initialization, manipulation, and readout of valley states. This is because they possess multiple inequivalent valleys in their band structures that do not couple distinctly to external optical fields. For example, as shown in Figure. 1.1(b), the dashed lines represent the band structure of unstrained silicon, which exhibits six equivalent indirect valleys located along high-symmetry directions.

The situation changed with the discovery of graphene in 2004, which triggered rapid progress in research on two-dimensional (2D) materials. Graphene has a hexagonal honeycomb lattice that places two inequivalent valleys at the K and K' points of the hexagonal Brillouin zone. Nevertheless, the absence of a bandgap in graphene, as shown in Figure. 1.1(d), allows electrons to be thermally excited between valleys, making valley polarization unstable at room temperature.

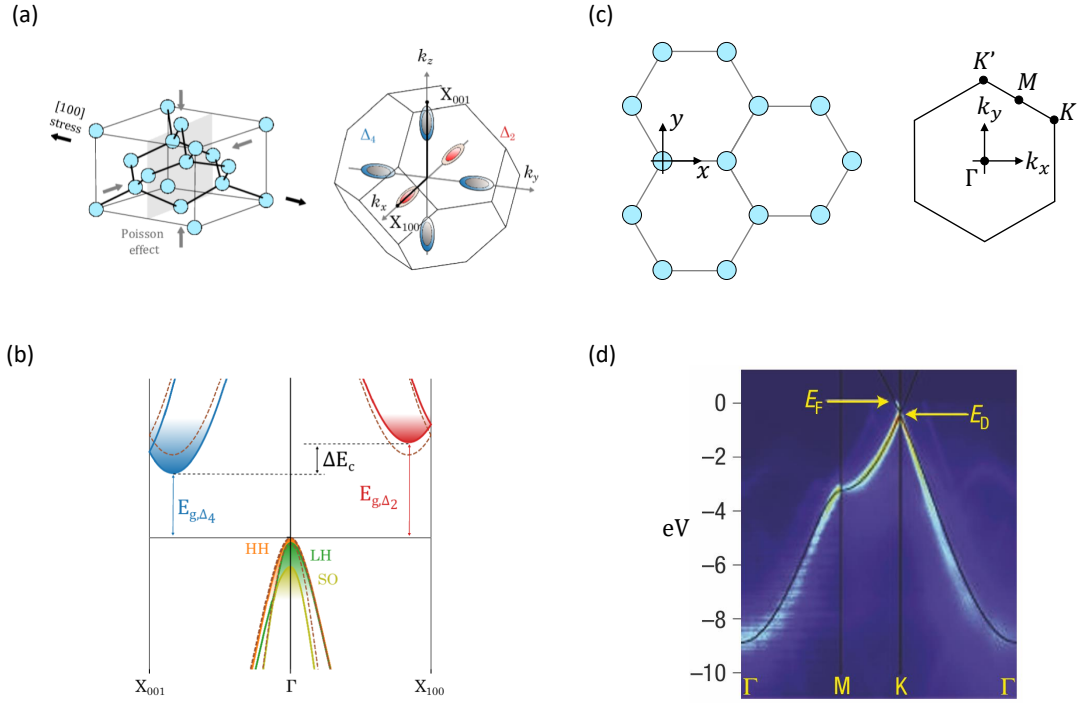


Figure 1.1 : (a) Illustration of the silicon lattice structure and Brillouin zone reproduced from Ref. [1]. (b) Calculated band structure under uniaxial tensile strain reproduced from Ref. [1]. (c) Illustration of the graphene lattice structure and Brillouin zone. (d) Graphene band structure measured using angle-resolved photoemission spectroscopy, reproduced from Ref. [2].

In contrast, transition-metal dichalcogenide (TMD) monolayers share a similar honeycomb lattice but possess a finite direct bandgap (see Figure. 1.2) on the order of $\sim 1\text{--}2$, eV, which lies in the visible to near-infrared spectral range. A TMD monolayer consists of two group-VIA chalcogen atoms, such as sulfur (S) or selenium (Se), with a group-VIB transition-metal atom, such as molybdenum (Mo) or tungsten (W), located between them.

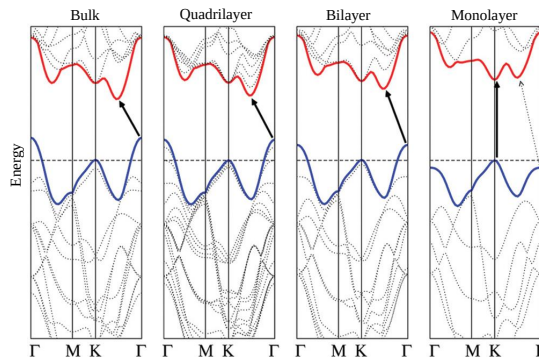


Figure 1.2 : Calculated band structures of MoS₂ with different layers, reproduced from Ref. [3].

By optically pumping a TMD monolayer, electrons can be excited from the valence band to the conduction band around the K and K' valleys, leaving holes in the valence band. Due to

the strong Coulomb interaction in atomically thin materials with weak dielectric screening, the electron-hole pair can bind together to form a quasiparticle known as an exciton.

Moreover, the broken inversion symmetry of the lattice, combined with strong spin-orbit coupling from the heavy transition-metal atoms, leads to spin splitting of the bands. The splitting at the K/K' valleys is typically on the order of a few meV in the conduction band and hundreds of meV in the valence band, as shown in Figure. 1.3. In addition, the combination of broken inversion symmetry and spin-orbit coupling results in valley-dependent optical selection rules: optical transitions at the K and K' valleys couple selectively to σ^+ and σ^- circularly polarized light. Such a band structure also constrains whether an exciton can be optically created or annihilated through photon absorption or emission, giving rise to bright excitons (BX) and dark excitons (DX). This property enables optical initialization and detection of valley polarization in monolayer TMDs.

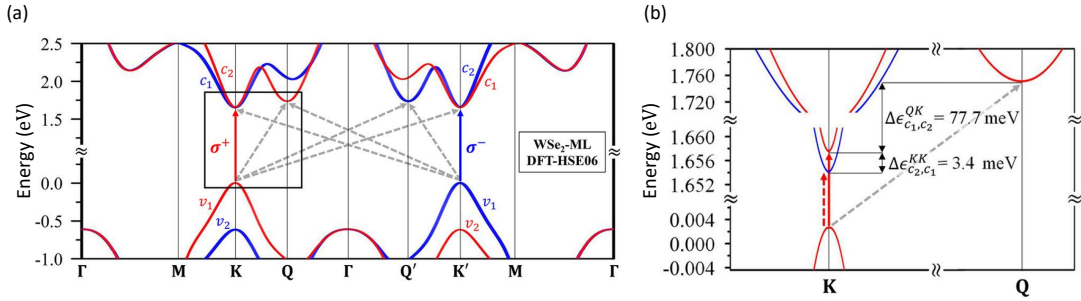


Figure 1.3 : Calculated band structures of WSe₂ with spin-up (red) and spin-down (blue) states reproduced from Ref. [4].

Furthermore, excitons can be categorized according to whether the momentum and optical selection rules are satisfied (see Figure. 1.4). On one hand, the center-of-mass momentum $\mathbf{Q}$ of a bright exciton (BX) must be smaller than or comparable to the photon momentum $\mathbf{Q}_c$, where the photon dispersion relation is $E = \hbar c |\mathbf{Q}_c|$. Bright excitons are composed of electron and hole states with antiparallel spin orientations and can couple efficiently to light. Only excitons with center-of-mass momentum inside the light cone can recombine radiatively.

On the other hand, dark excitons (DX) cannot couple directly to light due to either spin-forbidden or momentum-forbidden transitions. In W-based TMDs, an additional excitonic state known as the gray exciton (GX) can appear. Gray excitons emit light polarized perpendicular to the two-dimensional plane. Compared with BXs and DXs, GXs have attracted significant attention because of their relatively long lifetimes and weak but finite optical activity.

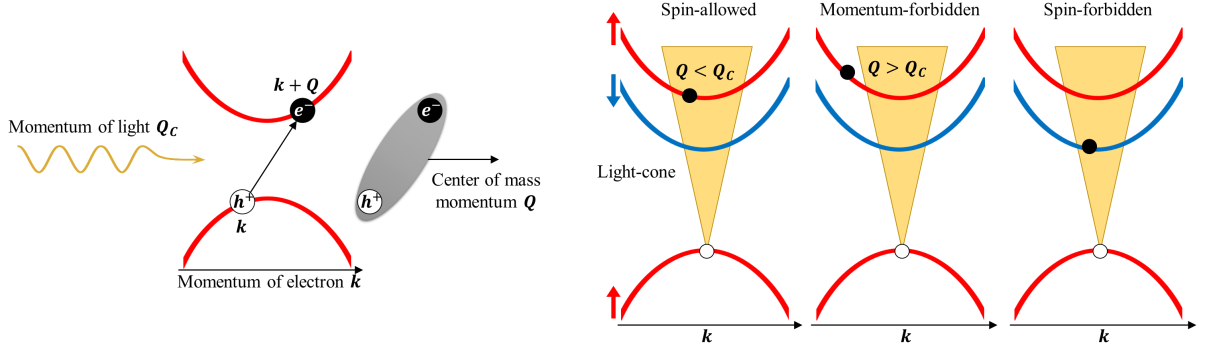


Figure 1.4 : Illustration of a photon carrying momentum Q_c exciting (leaving) an electron (hole) with momentum $k + Q$ ($-k$), forming an exciton with center-of-mass momentum Q . The configurations of spin-allowed bright excitons, momentum-forbidden dark excitons, and spin-forbidden dark excitons are also illustrated.

Many intriguing excitonic properties remain to be explored in two-dimensional TMD materials, offering opportunities for novel applications in valleytronics, optoelectronics, and quantum technologies.

1.2 Light Carrying Orbital Angular Momentum

Through the theoretical work of Poynting in 1909 [5] and the experimental demonstration by Beth in 1936 [6], it has been established that each photon in circularly polarized light carries a spin angular momentum (SAM) of $S = s\hbar = \pm\hbar$, where s is the photon helicity and $\hbar$ is the reduced Planck constant. In Beth's famous experiment, he measured the torque exerted on a suspended half-wave plate as circularly polarized light passed through it, demonstrating that spin angular momentum can be transferred to a birefringent medium (e.g., calcite) during the propagation of the beam (see Figure. 1.5(a)).

In 1992, Allen *et al.* demonstrated that a laser beam in a Laguerre–Gaussian (LG) mode possesses a well-defined orbital angular momentum (OAM) of $L = \ell\hbar$ per photon, where $\ell \in \mathbb{Z}$ is the azimuthal mode index, commonly referred to as the topological charge [7] (see Figure. 1.5(b)). This discovery was demonstrated by converting an input Hermite–Gaussian mode, which carries zero OAM, into an LG mode carrying nonzero OAM, which subsequently induced a measurable torque on cylindrical lenses. This experiment revealed that light can also carry OAM associated with the spatial structure of its wavefront. Figure. 1.6 illustrates the helical (twisted) phase structure associated with a quantized amount of $\ell\hbar$ of OAM, while the intensified charge coupled device camera images show the characteristic donut-shaped intensity distribution.

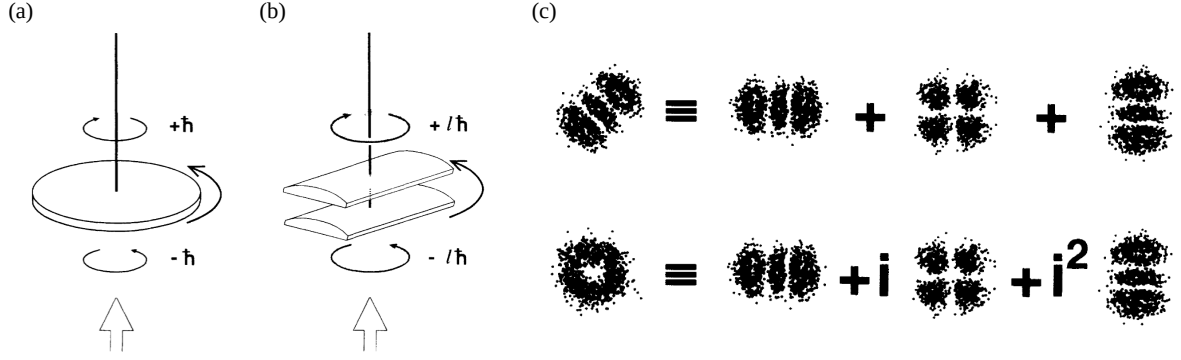


Figure 1.5 : (a) Illustration of the conversion between right-handed and left-handed circularly polarized light generating a torque on a suspended birefringent half-wave plate. (b) Illustration of transforming a Laguerre – Gaussian (LG) mode carrying OAM $\ell\hbar$ per photon into a mode with $-\ell\hbar$ per photon, which produces a mechanical torque on suspended cylindrical lenses. (c) Demonstration of generating an LG beam with nonzero ℓ (left side of bottom row) by introducing a phase shift when converting Hermite–Gaussian modes with zero ℓ (left side of top row). All reproduced from Ref. [7]

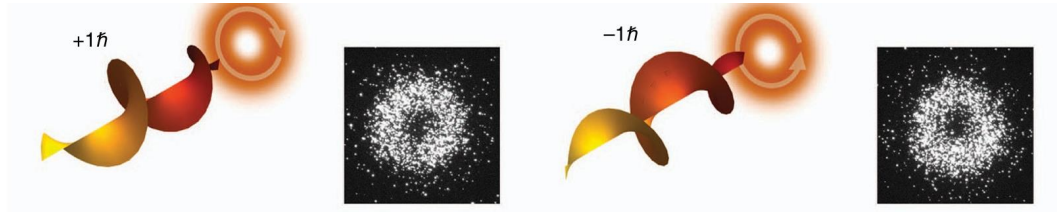


Figure 1.6 : Illustration of twisted light with a helical wavefront, where each photon carries $\ell\hbar$ of orbital angular momentum. The intensified charge coupled device camera images show the characteristic donut-shaped intensity profile. Reproduced from Ref. [8]

Since then, a wide range of applications of light carrying OAM have been developed. In optical communications [9–11], the additional degree of freedom provided by OAM, beyond wavelength, amplitude, phase, and polarization, can significantly enhance the data transmission capacity. For example, in 2013 a data transmission rate of 400 Gb/s was achieved using four OAM modes at a single wavelength in a 1.1 km-long optical fiber [11]. In 2016, long-distance free-space communication using twisted light over a distance of 143 km between two islands was successfully demonstrated (see Figure. 1.7(a–c)).

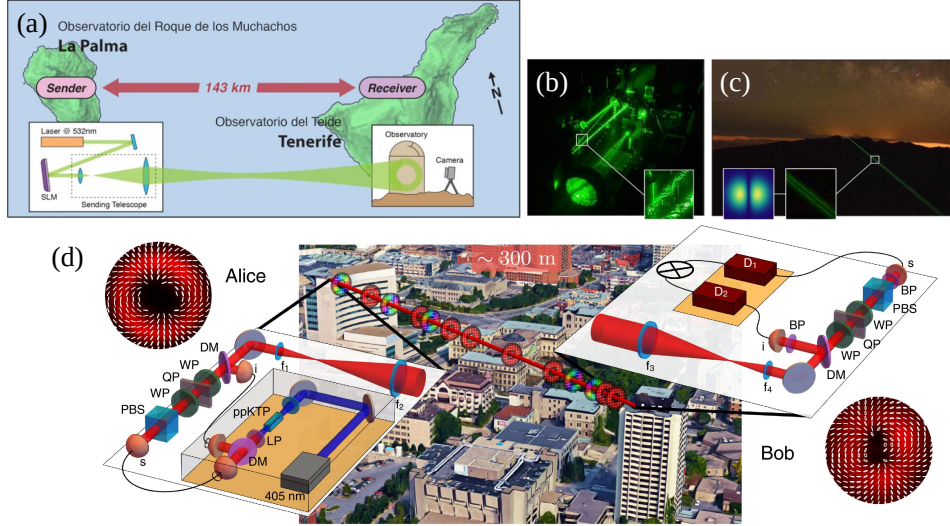


Figure 1.7 : (a) Schematic of optical communication between two islands separated by 143 kilometers. (b) Photograph of the transmitter. (c) Long-exposure photograph showing the superposition of twisted light beams. All reproduced from Ref. [12]. (d) Illustration of quantum key distribution between two buildings in Ottawa reproduced from Ref. [12]

The OAM of light also provides an additional degree of freedom for quantum key distribution [12, 13]. In 2017, a free-space communication link between the rooftops of two buildings separated by 300 meters in Ottawa was successfully demonstrated with an acceptable error rate while achieving a secret key rate approximately 50% higher than conventional methods [12] (see Figure. 1.7(d)).

In astronomy, the use of OAM modes in telescope systems has been proposed as a method to enable direct observational demonstrations of rotating black holes. Other applications include data storage [14, 15], enhanced sensitivity in imaging [16, 17], and optical tweezers [18, 19]. The possibility of utilizing OAM in twisted light has therefore opened new directions for both fundamental research and technological applications.

1.3 Laguerre–Gaussian (LG) beam in TMD

The topological charge characterizes the helical phase wavefront and the donut-shaped transverse intensity distribution of an LG beam, providing new possibilities for manipulating microscopic objects. In 2001, L. Paterson *et al.* demonstrated that LG beams with nonzero ℓ can be used to trap and rotate microscopic objects such as silica spheres, glass rods, and chromosomes [20], as illustrated in Figure. 1.8. The ability of twisted light to influence microscopic objects with OAM raises an important question: can orbital angular momentum of light also influence electronic excitations in condensed matter systems, such as excitons in 2D TMD?

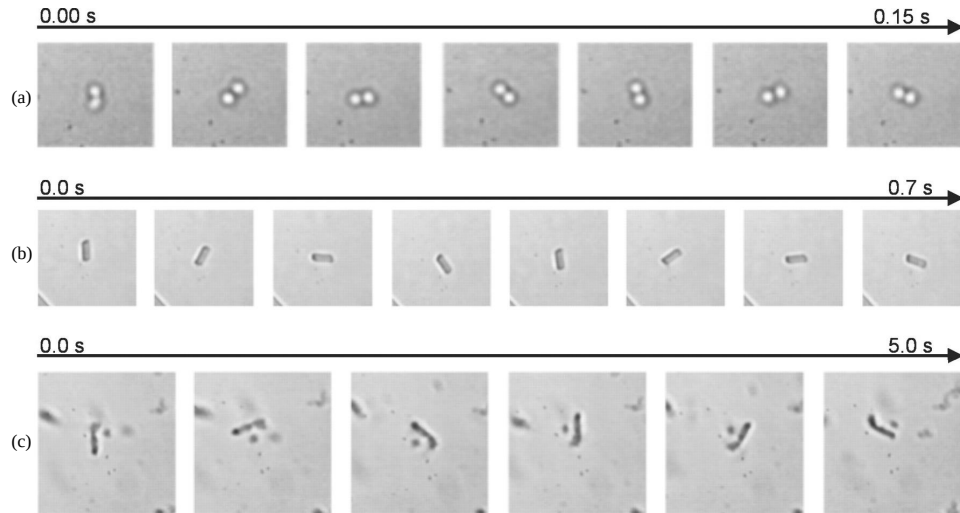
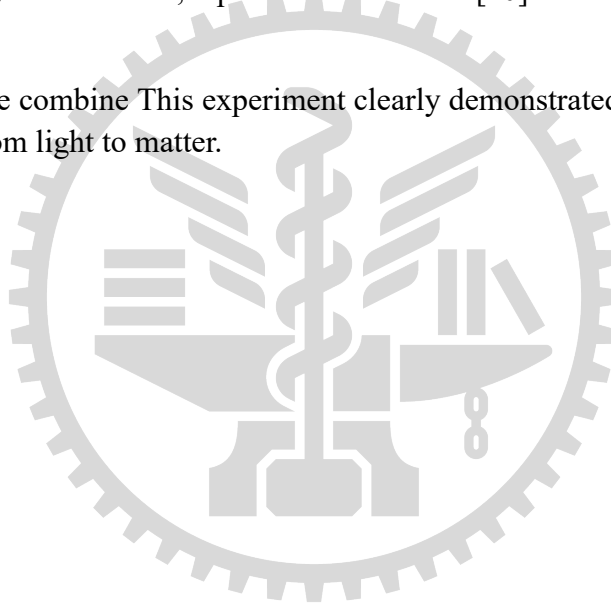


Figure 1.8 : Images of two-dimensional rotation of optically trapped objects: (a) silica spheres, (b) glass rod, and (c) chromosome, reproduced from Ref. [20]

In this work, we combine This experiment clearly demonstrated the transfer of orbital angular momentum from light to matter.



Chapter 2 Twisted-Light with Orbital Angular Momentum

2.1 Paraxial Helmholtz equation

2.1.1 Wave Equations

The Maxwell's equations describe how **electric field** $E(\mathbf{r}, t)$ and **magnetic field** $B(\mathbf{r}, t)$ behave in given position $\mathbf{r}$ and time t , including Laguerre-Gaussian beam. In **free space**, the charge density ρ and current density $\mathbf{J}$ are absent, making the Maxwell's equations four homogeneous differential equations with vacuum permittivity ϵ_0 and vacuum permeability μ_0 and speed of light in vacuum $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$. In electrodynamics scheme, the Maxwell's equations are

$$\nabla \cdot \mathbf{E}(\mathbf{r}, t) = \frac{\rho}{\epsilon_0} = 0, \quad (2.1)$$

$$\nabla \cdot \mathbf{B}(\mathbf{r}, t) = 0, \quad (2.2)$$

$$\nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial}{\partial t} \mathbf{B}(\mathbf{r}, t), \quad (2.3)$$

$$\nabla \times \mathbf{B}(\mathbf{r}, t) = \mu_0 \mathbf{J} + \frac{1}{c^2} \frac{\partial}{\partial t} \mathbf{E}(\mathbf{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \mathbf{E}(\mathbf{r}, t). \quad (2.4)$$

Where $\nabla \equiv \frac{\partial}{\partial x} \hat{\mathbf{x}} + \frac{\partial}{\partial y} \hat{\mathbf{y}} + \frac{\partial}{\partial z} \hat{\mathbf{z}}$ is the partial derivatives along each of the three coordinate directions. In addition, the four equations can be generalized in potential form and will be used in the light-matter interaction theory latter. First, the divergenceless of Eq. (2.2) introduce a **vector potential** $\mathbf{A}(\mathbf{r}, t)$:

$$\mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t). \quad (2.5)$$

Second, the presence of $\mathbf{A}(\mathbf{r}, t)$ and the Faraday's law (Eq. (2.3)) gives a quantity that vanishes with its curl and gives rise to a **scalar potential** $\phi(\mathbf{r}, t)$:

$$\nabla \phi(\mathbf{r}, t) = \left(\mathbf{E}(\mathbf{r}, t) + \frac{\partial}{\partial t} \mathbf{A}(\mathbf{r}, t) \right). \quad (2.6)$$

Then, in terms of $\mathbf{A}(\mathbf{r}, t)$ and $\phi(\mathbf{r}, t)$, then the electric field can be written as

$$\mathbf{E}(\mathbf{r}, t) = -\nabla \phi(\mathbf{r}, t) - \frac{\partial}{\partial t} \mathbf{A}(\mathbf{r}, t) \quad (2.7)$$

which will certainly satisfy both Eq. (2.2) and Faraday's law (Eq. (2.3)). Finally, the Gauss's law (Eq. (2.1)) and the Ampère's law with Maxwell's correction (Eq. (2.4)) can be rewritten as

two wave equations for the electromagnetic potentials

$$\nabla^2 \phi(\mathbf{r}, t) + \frac{\partial}{\partial t} (\nabla \cdot \mathbf{A}(\mathbf{r}, t)) = 0, \text{ and} \quad (2.8)$$

$$\nabla^2 \mathbf{A}(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}(\mathbf{r}, t) - \nabla \left(\nabla \cdot \mathbf{A}(\mathbf{r}, t) + \frac{1}{c^2} \frac{\partial}{\partial t} \phi(\mathbf{r}, t) \right) = 0. \quad (2.9)$$

The two equation remain all the information in Maxwell's equations.

2.1.2 Gauge Freedom

Further simplification of the wave equations Eq. (2.8) and Eq. (2.9) can be achieved by exploiting the gauge freedom of the vector and scalar potentials, $\mathbf{A}(\mathbf{r}, t)$ and $\phi(\mathbf{r}, t)$. Below, we introduce two commonly used gauges: the **Coulomb gauge** and the **Lorenz gauge**. The former is widely adopted in light-matter interaction theory, while the latter is employed here for the derivation of Laguerre-Gaussian beams. Superscripts C and L are used in the following derivations to distinguish quantities defined in the Coulomb and Lorenz gauges, respectively. The gauges conditions are given as

$$\text{Coulomb gauge condition : } \nabla \cdot \mathbf{A}^C(\mathbf{r}, t) = 0, \quad (2.10)$$

$$\text{Lorenz gauge conditon : } \nabla \cdot \mathbf{A}^L(\mathbf{r}, t) = -\frac{1}{c^2} \frac{\partial}{\partial t} \phi^L(\mathbf{r}, t). \quad (2.11)$$

Both gauges can simplify the wave equations Eq. (2.8) and Eq. (2.9). In Coulomb gauge, the wave equations turn into

$$\nabla^2 \phi^C(\mathbf{r}, t) = 0, \text{ and} \quad (2.12)$$

$$\nabla^2 \mathbf{A}^C(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}^C(\mathbf{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \nabla \phi^C(\mathbf{r}, t). \quad (2.13)$$

Where Eq. (2.12) has trivial solution $\phi^C(\mathbf{r}, t) = 0$ and Eq. (2.13) can be rewritten as $\nabla^2 \mathbf{A}^C(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}^C(\mathbf{r}, t) = 0$. In Lorenz gauge, the wave equations Eq. (2.8) and Eq. (2.9) turn into

$$\nabla^2 \phi^L(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \phi^L(\mathbf{r}, t) = 0, \text{ and} \quad (2.14)$$

$$\nabla^2 \mathbf{A}^L(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}^L(\mathbf{r}, t) = 0. \quad (2.15)$$

The physical quantity $\mathbf{E}(\mathbf{r}, t)$ can be related to vector potentials in different gauges through the relation

$$\mathbf{E}(\mathbf{r}) = i\omega \mathbf{A}^C(\mathbf{r}) = i\omega \left(\mathbf{A}^L(\mathbf{r}) + \frac{\nabla (\nabla \cdot \mathbf{A}^L(\mathbf{r}))}{q_0^2} \right) \quad (2.16)$$

The derivation of this relation has adopt the ansatz of **monochromatic wave** (for electric field $\mathbf{E}(\mathbf{r}, t) = \text{Re} [\mathbf{E}(\mathbf{r}) e^{-i\omega t}]$, scalar potentials $\phi^L(\mathbf{r}, t) = \text{Re} [\phi^L(\mathbf{r}) e^{-i\omega t}]$, vector potential $\mathbf{A}^C(\mathbf{r}, t) = \text{Re} [\mathbf{A}^C(\mathbf{r}) e^{-i\omega t}]$, and $\mathbf{A}^L(\mathbf{r}, t) = \text{Re} [\mathbf{A}^L(\mathbf{r}) e^{-i\omega t}]$ with ω angular frequency) and derived from Eq. (2.7) under corresponding gauge condition. Where $q_0 = \omega/c$ is wavenumber of light

2.1.3 Paraxial Helmholtz Equation

In principle, to obtain the solution in vector potential form we can keep on the derivation with either Eq. (2.13) or Eq. (2.15), in conclusion we will use **Lorenz gauge** with Eq. (2.15) and leave the reason at the end of this section. As the first step, the **monochromatic wave** ansatz ($\mathbf{A}^L(\mathbf{r}, t) = \text{Re} [\mathbf{A}^L(\mathbf{r})e^{-i\omega t}]$) turns Eq. (2.15) into vector Helmholtz equation

$$\nabla^2 \mathbf{A}^L(\mathbf{r}) + q_0^2 \mathbf{A}^L(\mathbf{r}) = 0. \quad (2.17)$$

Next, the **transverse wave** ansatz ($\mathbf{A}^L(\mathbf{r}) = \hat{\mathbf{e}} A^L(\mathbf{r})$ with polarization vector of unit modulus $\hat{\mathbf{e}}$) can be rewritten Eq. (2.17) as scalar Helmholtz equation

$$\nabla^2 A^L(\mathbf{r}) + q_0^2 A^L(\mathbf{r}) = 0. \quad (2.18)$$

We further assume that the rays have only small inclinations with respect to the optical axis, a condition referred to as the **paraxial approximation** in geometric optics. In wave optics, this corresponds to the angular spectrum of a paraxial light beam having dominant plane-wave components lie close to the beam propagation direction, which is chosen to be along the z axis. As a result, the largest component of a wavevector $\mathbf{q}_0$ lies along the $\hat{z}$ direction with

$$q_z = \sqrt{q_0^2 - q_{\parallel}^2} = q_0 \sqrt{1 - \frac{q_{\parallel}^2}{q_0^2}} \approx q_0 - \frac{q_{\parallel}^2}{2q_0}, \quad (2.19)$$

where the value of in-plane wave vector $q_{\parallel} = \sqrt{q_x^2 + q_y^2}$. With the approximation, we consider the **paraxial wave** ansatz ($A^L(\mathbf{r}) = U(\mathbf{r})e^{iq_0 z}$) and substitute into scalar Helmholtz equation (Eq. (2.18)) gives

$$\nabla_{\parallel}^2 U(\mathbf{r}) + \frac{\partial^2}{\partial z^2} U(\mathbf{r}) + 2iq_0 \frac{\partial}{\partial z} U(\mathbf{r}) = 0 \quad (2.20)$$

Where $\nabla_{\parallel}^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is the transverse Laplace operator. The ansatz features a slowly varying spatial distribution ($U(\mathbf{r})$) along z direction comparing to primary variation ($e^{iq_0 z}$), thus holds the following relations $\left| \frac{\partial^2}{\partial z^2} U(\mathbf{r}) \right| \ll \left| \nabla_{\parallel}^2 U(\mathbf{r}) \right|$, and $\left| \frac{\partial^2}{\partial z^2} U(\mathbf{r}) \right| \ll q_0 \left| \frac{\partial}{\partial z} U(\mathbf{r}) \right|$. Neglecting the second-order derivative with respect to z constitutes the paraxial approximation and results in the paraxial Helmholtz equation

$$\nabla_{\parallel}^2 U(\mathbf{r}) + 2iq_0 \frac{\partial}{\partial z} U(\mathbf{r}) = 0. \quad (2.21)$$

Note that paraxial wave ansatz will not hold the relations in Coulomb gauge since the gauge condition (Eq. (2.10)) gives

$$\nabla \cdot \mathbf{A}^C(\mathbf{r}, t) = \hat{\mathbf{e}} \cdot \nabla A^C(\mathbf{r}, t) = 0. \quad (2.22)$$

For a transverse wave whose polarization vector only have x and y components, Eq. (2.22) simplifies to $\hat{\mathbf{e}} \cdot \nabla_{\parallel} A^C(\mathbf{r}, t) = 0$, which forces zero transverse variation along the in-plane direction ($\nabla_{\parallel} = \frac{\partial}{\partial x} \hat{\mathbf{x}} + \frac{\partial}{\partial y} \hat{\mathbf{y}}$). Thus violate our ansatz that $U(\mathbf{r})$ is slowly varying along z direction, unless a non-zero z component is allowed to exist to satisfy the divergence free condition.

2.2 Twisted Light in Laguerre-Gaussian mode

2.2.1 Real space representation under the Lorenz gauge

The paraxial Helmholtz equation (Eq. (2.21)) can be solved under various coordinate systems and yields different solutions such as **Hermite-Gaussian** beam, **Laguerre-Gaussian** (LG) beam, and **Bessel** beams, etc. In the twisted Light theory we take the widely used mode, LG beams, solved from the paraxial Helmholtz equation in cylindrical coordinates

$$\left(\frac{1}{\rho} \frac{\partial}{\partial \rho} + \frac{\partial^2}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial^2}{\partial \phi^2} + 2iq \frac{\partial}{\partial z} \right) U(\rho, \phi, z) = 0 \quad (2.23)$$

Where the position vector $\mathbf{r} = (x, y, z) = (\mathbf{r}_{\parallel}, z) = (\rho, \phi, z)$, and $\mathbf{r}_{\parallel}$ is the in-plane position vector which contain ρ radial coordinate and ϕ azimuthal angle. By considering **long Rayleigh-length conditionn**, the LG beam is than solved as

$$\mathbf{A}_{\mathbf{q}_0}^{\hat{\epsilon}, \ell, p; L}(\mathbf{r}) = \hat{\epsilon} A_0^{\text{LG}} u_{\ell p}(\mathbf{r}_{\parallel}) e^{iq_0 z}, \quad (2.24)$$

where the subscripted integer $\ell \in \mathbb{Z}$ is the index of azimuthal mode and refer as **topological charge**, subscripted integer $p \geq 0$ is the index of radial mode, $\mathbf{q}_0 = q_0 \hat{z}$ is the wavevector specified to be along the direction of light propagation, and A_0^{LG} is the constant amplitude of the vector potential. The spatial distribution is expressed as

$$U(\rho, \phi, z) \approx u_{\ell p}(\mathbf{r}_{\parallel}) = f_{\ell p}(\rho) e^{i\ell \phi}, \quad (2.25)$$

and follows

$$f_{\ell p}(\rho) = \sqrt{\frac{2p!}{\pi(|\ell| + p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} \exp \left(-\frac{\rho^2}{W_0^2} \right), \quad (2.26)$$

where $L_p^{|\ell|}(x) = \sum_{m=0}^p (-1)^m [(|\ell| + p)! / (p - m)! (|\ell| + m)! m!] x^m$ is the **generalized Laguerre polynomial** with the variable $x = \left(\frac{2\rho^2}{W_0^2} \right)$, and parametrized by the beam waist W_0 , $\rho = \sqrt{x^2 + y^2}$, $\phi = \tan^{-1}(y/x)$.

2.2.2 Angular spectrum representation under the Lorenz gauge

Since the light-matter interaction theory include LG beams in angular spectrum representation, we need to describe the vector potential in terms of plane-wave basis

$$\mathbf{A}(\mathbf{r}) = \sum_{\mathbf{q}} \mathcal{A}(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{r}}, \quad (2.27)$$

where $\mathcal{A}^L(\mathbf{q})$ is the (complex) amplitude with each $e^{i\mathbf{q}\cdot\mathbf{r}}$ as plane-wave basis. The LG beam in Eq. (2.24) can obtain this form through **two-dimensional Fourier transformation**

$$u_{\ell p}(\mathbf{r}_{\parallel}) = \frac{1}{(2\pi)^2} \int d^2\mathbf{q}_{\parallel} F_{\ell p}^{\text{LG}}(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel}\cdot\mathbf{r}_{\parallel}}, \quad (2.28)$$

and determine $\mathcal{A}^L(\mathbf{q}_{\parallel})$ with the **inverse transformation**

$$F_{\ell p}^{\text{LG}}(\mathbf{q}_{\parallel}) = \int d^2\mathbf{r}_{\parallel} u_{\ell p}(\mathbf{r}_{\parallel}) e^{-i\mathbf{q}_{\parallel}\cdot\mathbf{r}_{\parallel}}, \quad (2.29)$$

where the in-plane wave vector $\mathbf{q}_{\parallel} = (q_{\parallel}, \phi_{q_{\parallel}})$. Since the spatial distribution has the form $u_{\ell p}(\mathbf{r}_{\parallel}) = f_{\ell p}(\rho) e^{i\ell\phi}$, the amplitude for plane-wave basis can be solved analytically as $F_{\ell p}^{\text{LG}}(\mathbf{q}_{\parallel}) = (-1)^{|\ell|} 2\pi \mathcal{H}_{|\ell|}[f_{\ell p}(\rho)] e^{i\ell\phi_{q_{\parallel}}}$, where the **Hankel transformation** $\mathcal{H}_{|\ell|}[f_{\ell p}(\rho)] = \int_0^\infty d\rho \rho f_{\ell p}(\rho) J_{|\ell|}(q_{\parallel}\rho)$ of order $|\ell|$ and $J_{|\ell|}(q_{\parallel}\rho)$, the **Bessel function of the first kind** of order $|\ell|$. Further approximately regard $\mathbf{q}_{\parallel}$ as the vector for 2D material in reciprocal space so that Eq. (2.28) can be discretized and the LG beams in angular spectrum representation become

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p; L}(\mathbf{r}) = \hat{\varepsilon} \sum_{\mathbf{q}_{\parallel}} \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel}\cdot\mathbf{r}_{\parallel}} e^{iq_0 z}, \quad (2.30)$$

where the amplitude function of LG beam is given by

$$\mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) = \tilde{F}_{\ell p}(q_{\parallel}) e^{i\ell\phi_{q_{\parallel}}}, \quad (2.31)$$

and the radial function $\tilde{F}_{\ell p}(q_{\parallel}) = \frac{A_0^{\text{LG}}}{\Omega} (-1)^{|\ell|} 2\pi \mathcal{H}_{|\ell|}[f_{\ell p}(\rho)]$. This study focuses on the fundamental radial mode $p = 0$ for LG beams, which have explicit form in real space as

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; L}(\mathbf{r}) = \hat{\varepsilon} \frac{A_0^{\text{LG}}}{\sqrt{|\ell|!} \pi/2} \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} \exp\left(-\frac{\rho^2}{W_0^2}\right) \exp[i(\ell\phi + q_0 z)], \quad (2.32)$$

whereas in angular spectrum representation

$$\mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) = \left(\frac{\pi A_0^{\text{LG}} W_0^2 (-1)^{|\ell|}}{\Omega \sqrt{|\ell|!} \pi/2} \right) \left[\left(\frac{q_{\parallel} W_0}{\sqrt{2}} \right)^{|\ell|} \exp\left(-\frac{q_{\parallel}^2 W_0^2}{4}\right) \right] \exp(i\ell\phi_{q_{\parallel}}) \quad (2.33)$$

add some figure

2.2.3 Real space representation under the Coulomb gauge

While the light-matter interaction theory include LG beams in Coulomb gauge, The LG beams are derived in Lorenz gauge. Therefore, it is vital to convert our solutions into Coulomb gauge. In previous section we already have Eq. (2.16) that implicit the connection between

vector potentials in different gauges, which gives us

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p; C}(\mathbf{r}) = \mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p; L}(\mathbf{r}) + \frac{\nabla (\nabla \cdot \mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p; L}(\mathbf{r}))}{q_0^2}. \quad (2.34)$$

For LG beams in fundamental radial mode $p = 0$

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; C}(\mathbf{r}) \approx A_0^{\text{LG}} \left[f_{\ell p=0}(\rho) e^{i\ell\phi} \hat{\varepsilon} + \frac{i}{\sqrt{2}q_0\rho} \left((|\ell| + \sigma\ell) - \frac{2\rho^2}{W_0^2} \right) f_{\ell p=0}(\rho) e^{i(\sigma+\ell)\phi} \hat{z} \right] e^{iq_0 z} \quad (2.35)$$

$$\hat{\varepsilon} = \frac{1}{\sqrt{2}}(\hat{x} + i\sigma\hat{y})$$

2.2.4 Angular spectrum representation under the Coulomb gauge

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; C}(\mathbf{r}) = \frac{1}{(2\pi)^2} \int d^2\mathbf{q}_{\parallel} \mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z} \quad (2.36)$$

$$\mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel}) = \left[\hat{\varepsilon} - \left(\frac{|\hat{\mathbf{q}}||\hat{\mathbf{q}}_{\parallel}|}{q_0^2} \right) (\hat{\varepsilon} \cdot \hat{\mathbf{q}}_{\parallel}) \hat{\mathbf{q}} \right] \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) \quad (2.37)$$

$$= \left[\hat{\varepsilon} - \left(\frac{|\mathbf{q}|}{q_0} \right)^2 \frac{\sin\theta_{q_{\parallel}}}{\sqrt{2}} e^{i\sigma\phi_{q_{\parallel}}} \hat{\mathbf{q}} \right] \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) \quad (2.38)$$

$$\approx \left[\hat{\varepsilon} - \frac{\sin\theta_{q_{\parallel}}}{\sqrt{2}} e^{i\sigma\phi_{q_{\parallel}}} \hat{\mathbf{q}} \right] \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) \quad (2.39)$$

$$(2.40)$$

$$\mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel}) \approx \tilde{F}_{\ell p}(q_{\parallel}) e^{i\ell\phi_{q_{\parallel}}} \hat{\varepsilon} - \frac{\sin\theta_{q_{\parallel}}}{\sqrt{2}} \tilde{F}_{\ell p}(q_{\parallel}) e^{i(\sigma+\ell)\phi_{q_{\parallel}}} \hat{z} \quad (2.41)$$

2.2.5 Vector Vortex Beams

Chapter 3 Light-Matter Interaction

3.1 Optical Transition with Single Twisted Light

3.1.1 Optical Transition with Single Twisted Light

The transition rate of an exciton state excited by time-dependent perturbation theory, the time-dependent exciton state under a weak photoexcitation can be expressed as

3.2 Optical Transition with Vector Vortex Beam



Chapter 4 Summary and outlook



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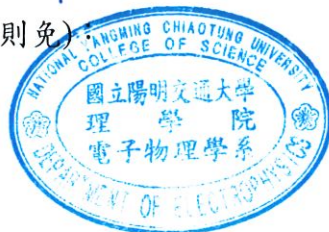
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